

Lactic Acid Spectroscopy: Intra- and Intermolecular Interactions

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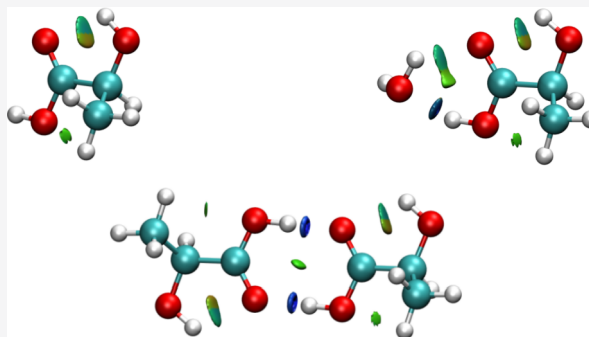
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ABSTRACT: Lactic acid, a relevant molecule in biology and the environment, is an α -hydroxy acid with a high propensity to form hydrogen bonds, both internally and to other hydrogen-bond-accepting molecules. This work includes the novel recording of infrared spectra of gas-phase lactic acid using Fourier transform infrared spectroscopy, and the vibrational absorption features of lactic acid are assigned with the aid of computationally simulated vibrational spectra with anharmonic corrections. Theoretical chemistry methods are used to relate intramolecular hydrogen-bond strengths to the relative stability of lactic acid conformers. The formation of hydrogen-bonded lactic acid dimers and 1:1 water complexes is investigated by simulated vibrational spectra and calculated thermodynamic parameters for the lactic acid monomer and dimer and its water complex in the gas phase. The results of this study are discussed in the context of environmental chemistry with an emphasis on indoor environments.



INTRODUCTION

The focal point of this work is the gas-phase vibrational spectroscopy of lactic acid and its ability to form intra- and intermolecular hydrogen bonds. Lactic acid, a short-chain α -hydroxy acid, is a ubiquitous biomolecule in both indoor and atmospheric settings.^{1–3} It is found in blood and sweat,⁴ with lactic acid evaporation off of human skin resulting in a marked increase in its concentration in indoor air.³ Lactic acid also has extensive uses in many industries as a food preservative, a flavoring agent, and a curing agent, among other things.⁵ It is used to synthesize polylactic acid, an inexpensive biodegradable polymer used in a variety of applications such as 3-D printing.^{6,7} In addition to human excretion, lactic acid can also be formed in the atmosphere through isoprene photo-oxidation.¹ In general, atmospheric organic acids are known to be aerosol precursors and have a rich atmospheric chemistry.^{8–11} Lactic acid specifically has been shown to enhance new particle formation.² The particulate matter content of air affects its overall quality, which has an important influence on human health.¹² Thus the study of lactic acid in the gas phase represents an important step forward in understanding the underlying processes governing lactic acid chemistry and could help future studies to investigate the consequences of elevated lactic acid concentrations in air.

Gas-phase infrared (IR) spectroscopy combined with theory is a synergistic mix of techniques for understanding molecular structures, characterizing hydrogen-bonded complexes,^{13,14} and elucidating intramolecular hydrogen-bonding effects.^{15–19} Theoretical chemistry methods are useful for spectroscopic

assignments, help with interpreting the structural and electrostatic effects on spectra, and can provide chemical properties of the studied molecule(s). Here we use Fourier transform infrared (FTIR) spectroscopy to study gas-phase lactic acid under light vacuum. This has, to the best of our knowledge, not been done before, likely due to the low vapor pressure preventing molecules from entering the gas phase.^{20,21} Density functional theory (DFT) is used to calculate anharmonically corrected vibrational spectra to corroborate assignments in the FTIR spectra. We also use single-point coupled-cluster calculations to obtain more accurate electronic energies for the molecules investigated.

In addition to probing the basic structure of gas-phase lactic acid, we explore both intramolecular hydrogen bonding in lactic acid and its ability to form intermolecular hydrogen bonds, specifically with water. In recent years, there have been new developments in applying electron density topology methods, which have been proven useful to gauge the strength of hydrogen bonds and accurately characterize them.^{13,22–26} Hydrogen bonding is naturally expected to have a major influence on the chemistry of lactic acid, as it is an α -hydroxy acid. Indeed, lactic acid is expected to form hydrogen bonds

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with water, other organic acids, or itself, including forming the cyclic lactic acid dimer. Thus the lactic acid monomer, its water complex, and dimer are studied computationally in this work, and the results are compared to experiments and relevant literature.

EXPERIMENTAL METHODS

L-Lactic acid (MP Biomedicals, >98% purity, referred to as “lactic acid” in this work) was used without further purification. Nuclear magnetic resonance spectrometry (^1H NMR, Bruker Fourier 300) was used to verify the purity of the lactic acid prior to spectra collection. A sample ^1H NMR spectrum is shown in the Supporting Information (SI), Figure S1. The sample chamber was a cross-shaped glass cell with an optical path length of 50 cm and a total volume of ~ 3 L. The horizontal portion of the sample cell was terminated on either end with CaF_2 windows. Prior to spectra collection, the sample was placed in the bottom of the vertical portion of the sample chamber, and the top was sealed to retain the vacuum. The windows and top were all sealed using Viton O-rings and custom PTFE clamps. A diagram of the chamber is provided in Figure S2. The pressure was monitored by a 10 Torr Baratron absolute pressure transducer. The lower portion of the sample chamber was wrapped with heating tape for temperature control, and the temperature was monitored by a type-K thermocouple attached to the outside of the sample cell, touching the glass of the cell and the heat tape. The procedure was to first evacuate the chamber to <0.2 Torr, then it was isolated, and a background spectrum was recorded. After the background spectra were recorded, ~ 1 g of lactic acid was added to the bottom of the cell. The cell was then again evacuated and pumped on for 30 min to remove as much residual contaminant, which was mainly water, as possible without pumping out too much lactic acid. The cell was then isolated, and lactic acid was allowed to equilibrate between the solid and gas phases for 10 min before spectrum recording took place. Finally, heat tape was used to heat the sample; then, spectra were recorded at different temperatures to allow more lactic acid to enter the gas phase. The IR absorption spectrum of lactic acid was measured between 1000 and 4000 cm^{-1} using FTIR with a Bruker IFS 66v/s instrument with an MCT detector, a Globar light source, and a KBr beamsplitter. The details of this setup have been previously described.^{11,27–30} Each FTIR spectrum consists of 200 scans with 2 cm^{-1} resolution. We used the OriginPro software³¹ for baseline smoothing and to obtain peak maxima and integrated absorbances of signals in the IR spectra by fitting Gaussian functions after having applied a baseline correction. Data visualization was done in Matlab.³² To quantify the amount of water in the experiments, the FTIR spectra were analyzed using the Matlab software, Main Polwin, which was developed at CEAM (Spain). It is based on a linear square fitting routine modified to improve the filtering process to remove baseline and broad absorptions of unknown compounds. This software compares standardized spectra, recorded at 0.5 cm^{-1} resolution, to the spectra in this work, recorded at 2 cm^{-1} resolution. The standardized spectra and spectra in this work were collected with the same apodization function (Happ-Genzel).

COMPUTATIONAL METHODS

We employed DFT calculations to identify the ground-state geometries of lactic acid, the associated harmonic frequencies for each minimum energy structure, and the thermodynamic properties³³ of each geometry. In this work, the B3LYP functional^{34,35} was used in combination with the may-cc-pVTZ basis set,^{36,37} and the calculations were carried out using the Gaussian 16 program revision A.03.³⁸ The B3LYP functional has previously been shown to provide relative energies between conformers of different molecules that are in good agreement with experiments and high-level theoretical methods.³⁹ Furthermore, our choice of B3LYP functional combined with the may-cc-pVTZ basis set was based on some initial testing of different basis sets as well as the $\omega\text{B97X-D}^{40}$ and B3LYP-D3^{34,35,41} functionals; see Section S10 of the SI for details. We take extra steps to validate the choice of method in this work, as presented in the Results and Discussion section and Sections S4 and S10 of the SI. These DFT calculations serve as the theoretical foundation for subsequently calculating vibrational transitions with anharmonic corrections and as methods for the analysis of electron density topology; see below. Single-point electronic energies were also calculated for all B3LYP/may-cc-pVTZ-optimized structures using the CCSD(T)/cc-pVTZ method^{42,43} in the CFOUR program.^{44,45} This increases confidence in the calculated relative energies for each conformer and complexation energies for lactic acid dimers and water complexes.

Theoretical Vibrational Spectra with Anharmonic Corrections. To calculate vibrational frequencies and intensities with anharmonicity included, vibrational second-order perturbation theory (VPT2) was used, as implemented in the Gaussian 16 program, on the optimized geometries.^{38,46–49} Part of the reason why the may-cc-pVTZ basis set was chosen in this work was that a triple- ζ basis set has been reported as the minimum acceptable basis set size when running VPT2 calculations in combination with DFT.⁵⁰ Although scaling factors are more commonly used to simulate anharmonicity, VPT2 was used to obtain anharmonically corrected vibrational intensities.⁵⁰ Each calculated transition was convoluted with a Gaussian function with a full width at half-maximum (fwhm) of 20 cm^{-1} . Convolution with a fixed fwhm value was done for visualization purposes, and 20 cm^{-1} fwhm was chosen as it approximately matches the width of the experimentally observed IR transitions; see Table S6 of the SI.

Electron Density Topology Analysis Methods. Topological analysis of the electron density of the specific structures was carried out to study hydrogen-bonding interactions with the Atoms in Molecules (AIM) theory^{22,51} and the Non-Covalent Interactions (NCI) index.²³ To do this, AIM2000⁵² and a locally developed NCI program named Bonder were used (Bonder code available upon request, contact Joseph Lane). By using the keyword “output = wfn” in the calculations, the Gaussian 16 program can generate files that AIM2000 and Bonder can interpret.³⁸ The AIM2000 program⁵² uses the electron density in the Cartesian space ($\rho(\mathbf{r})$) to identify points with a zero first derivative of the electron density ($\nabla\rho(\mathbf{r}) = 0$), also known as extrema, and then utilizes the second derivative ($\nabla^2\rho(\mathbf{r})$) evaluated at those points to determine what type of bond it is. The second derivative matrix, called a Hessian matrix, has three eigenvalues that are used to identify critical points. These are λ_1 , λ_2 , and λ_3 with the condition that $\lambda_1 < \lambda_2 < \lambda_3$. The sign of the

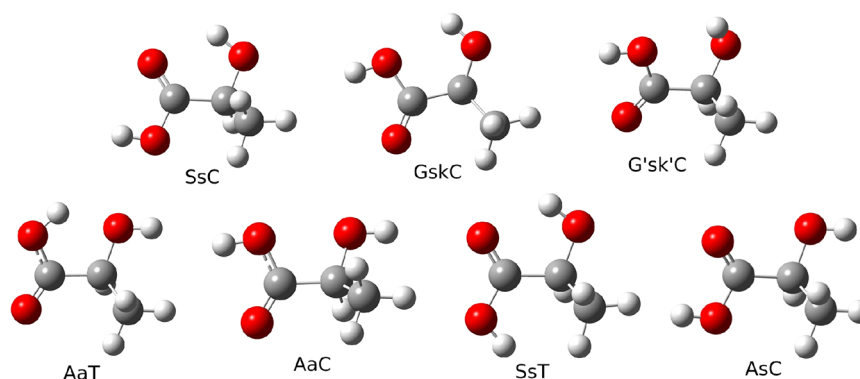


Figure 1. Identified lactic acid conformers sorted (left to right, top to bottom) by calculated relative stability.

eigenvalues characterizes the critical point. If all are negative, $\lambda_1 < 0$, $\lambda_2 < 0$, and $\lambda_3 < 0$, then the electron density decreases in all directions, which characterizes an atom, and this is useful as a consistency check. If one eigenvalue is positive, $\lambda_1 < 0$, $\lambda_2 < 0$, and $\lambda_3 > 0$, then it is a bond critical point (BCP). Topological analysis with AIM theory has been successfully used to correlate the electron density in BCPs (ρ_{BCP}) to bond strength, including hydrogen-bond strength.^{53,54} Often, hydrogen bonds will have an identifiable BCP with the AIM theory, but there are some notable exceptions.^{55,56} Methyl lactate, for example, is a molecule where AIM does not identify a BCP belonging to the internal hydrogen bond, whereas NCI clearly identifies a significant interaction.^{17,55} NCI analysis makes use of the reduced density gradient, $s(\mathbf{r})$, given by^{23,57}

$$s(\mathbf{r}) = \frac{|\nabla\rho(\mathbf{r})|}{2(3\pi^2)^{1/3}\rho(\mathbf{r})^{4/3}} \quad (1)$$

Typically, values of $s(\mathbf{r}) \leq 0.5$ are used to map out regions of interest when investigating hydrogen-bonding interactions. Most of these regions are naturally centered around the previously established BCPs, as $\Delta\rho(r) = 0$ means that $s(\mathbf{r}) = 0$. However, regions of low reduced density gradient can also occur without the presence of a BCP.⁵⁶ It is possible to calculate the kinetic energy density evaluated within an isovolume with the boundary condition set to $s(\mathbf{r}) = 0.5$, and this has been shown to be a good measure of the strength of the hydrogen-bonding interaction.^{13,18,19,25,26} The kinetic energy density of a hydrogen bond's $s(\mathbf{r}) = 0.5$ isosurface correlates strongly with the experimental red shift seen for bimolecular hydrogen-bonded complexes with alcohols as donor molecules and a variety of N-, O-, P-, and S-containing molecules as acceptors.¹³ The kinetic energy density is calculated directly from the electron density with the equation^{24,58}

$$G(\mathbf{r}) = \frac{3}{10}(3\pi^2)^{2/3}\rho(\mathbf{r})^{5/3} + \frac{[\nabla\rho(\mathbf{r})]^2}{72\rho(\mathbf{r})} + \frac{\nabla^2\rho(\mathbf{r})}{6} \quad (2)$$

In this work, we calculate the integrated kinetic energy density $G(s_{0.5})$ using Bonder, which allows the automatic identification of discrete NCI isosurfaces and the integration of various properties within each isosurface. Select calculations were repeated with the NCImilano²⁴ program to corroborate the results obtained with Bonder, the results of the two programs were in agreement.

RESULTS AND DISCUSSION

Lactic acid IR spectra from experimental and theoretical studies are presented here. The interpretation of experimental IR spectra is discussed in the context of these results. The monomeric 1:1 water complex and hydrogen-bonded dimer forms of lactic acid are explored in the sections to follow.

Lactic Acid Monomers and Vibrational Spectra. The structure and chemical properties of lactic acid were investigated computationally with DFT and experimentally using FTIR spectroscopy, as described in the [Experimental Methods](#) section. The combination of methods was chosen to identify and assign the vibrational spectroscopic features of gas-phase lactic acid.

Several different lactic acid conformers have previously been reported in the literature by Borba et al.,⁵⁹ who used DFT and Møller–Plesset second-order perturbation theory (MP2) to identify them, and the established naming convention^{17,60} that they used is repeated here; see [Figure 1](#) and [Section S3 of the SI](#) for details.

The Gibbs energy of each conformer was obtained through a combination of coupled-cluster and DFT calculations, as described in the [Computational Methods](#) section, and results were used to estimate the relative population of each conformer under the experimental conditions; see [Table 1](#). Relative Gibbs energies, ΔG , are calculated against the most

Table 1. Comparison of Relative Gibbs Energies (ΔG) and Relative Conformer Populations (pop.) between the Different Lactic Acid Conformers (conf.) Using Theoretical (This Work and Zeinalipour-Yazdi et al.⁶⁵) and Experimental Results (Borba et al.⁵⁹)

	ΔG (kJ/mol)		pop. (%)	
	this work	Yazdi et al. ⁶⁵ ^a	this work ^b	Borba et al. ⁵⁹ (Ar/Xe) ^c
SsC	0.0	0.0	94.0	95.5/96.9
GskC	8.5	9.3	2.8	3.6/2.2
G'sk'C	9.4	no data	2.0	no data ^d
AaT	10.8	9.9	1.1	0.9/0.8
AaC	18.0	17.4	0.06	no data
SsT	19.2	19.3	0.03	no data
AsC	20.0	18.6	0.02	no data

^aRelative Gibbs energy calculated at the B3LYP-D3/aug-cc-pVTZ level of theory. ^bCalculated at $T = 18^\circ\text{C}$. ^cMatrix isolation spectroscopy with population ratios given in argon/xenon matrices (Ar/Xe). ^dGskC and G'sk'C conformers were likely indistinguishable in their experiments.

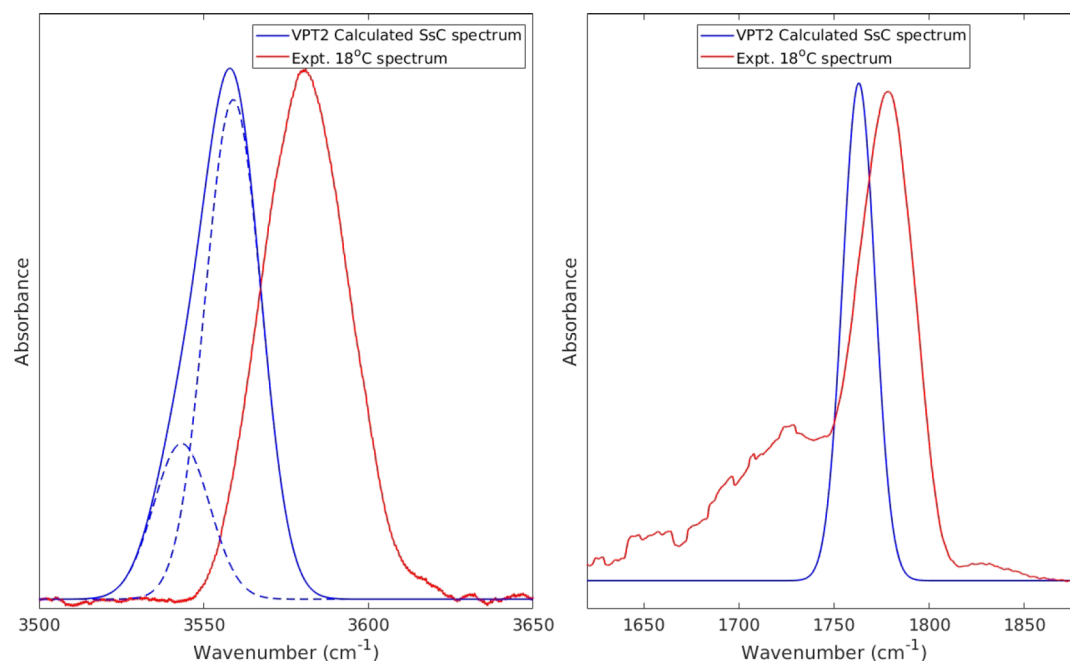


Figure 2. OH stretch (left) and C=O stretch (right) regions of the lactic acid infrared spectrum. The experimental spectrum (red) is compared with the simulated spectrum for the SsC conformer (blue, solid). Individual contributions to the total simulated spectrum in the OH stretch region are shown by the dashed blue lines.

stable conformer, SsC. These abundances are in good agreement with previous computational studies on lactic acid.^{59,61–65} These prior studies used a variety of methods, including Hartree–Fock (also known as SCF) calculations,⁶¹ MP2 with different basis sets,^{59,61,63–65} and DFT with different functional and basis set combinations.^{59,62}

The conformer distribution resembles that of methyl lactate (see Section S3 of the SI for its geometry), as the three lowest energy conformers, SsC, GskC, and G'sk'C, in lactic acid and methyl lactate are analogous.^{17,60} The IR spectrum was simulated for all lactic acid conformers according to the methods outlined previously and compared with experimental results. Given the relative populations shown in Table 1, the SsC conformer is expected to dominate the vibrational spectrum. In Figure 2, two IR regions with prominent lactic acid signals are shown, and select assignments of the calculated transitions to the experimental signals can be seen in Table 2. In Section S7 of the SI, we have included some additional

discussion about the assignments of the other signals that are observed, including the location of peak maxima and the integrated absorbances for each absorption feature.

There are several distinct signals belonging to lactic acid in the IR spectra. The most obvious ones are the OH band, the C=O stretch, and the ν_{16} skeletal mode signal. The peak placement of these and relative intensities are generally in good agreement with the corresponding calculated vibrational transitions; see Figure 2 and Table 2. The single OH band in the experimental IR spectra is assigned to contain both of the lactic acid OH stretching vibrational transitions (acid and alcohol). The total band has a full width at half-maximum of 26 cm^{-1} . The two fitted Gaussian functions to the experimental signal find the separation between the acid OH stretch and alcohol OH stretch to be 12 cm^{-1} , which lines up well with the simulated spectrum that predicts a 16 cm^{-1} separation between the two OH stretches. It can be seen from the simulated SsC spectrum in Figure 2 that the two OH signals are predicted to merge into one observed band, giving credence to assigning the OH band measured at 3580 cm^{-1} as a combination of both the acid and alcohol OH stretch transitions. There are several examples in the literature that support assigning the acid and alcohol OH stretches to one spectral feature. Matrix-isolation spectroscopy experiments done by Borba et al.⁵⁹ used argon as the host gas (at $T = 9$ K) to isolate lactic acid. Matrix-isolation IR spectroscopy often produces very narrow and sharp signals, allowing them to separate the two OH stretch transitions, but can also alter vibrational coupling patterns and relative intensities.^{66,67} Thus matrix-isolation IR spectroscopy is a good, but not exact, predictor for the experimental gas-phase spectrum. Borba et al.⁵⁹ observed a <4 cm^{-1} difference in peak maxima between the OH transitions,⁵⁹ which further supports a merging of the transitions to form one broad feature in the experimental spectrum. The vibrational transitions observed in glycolic acid may also serve as a good comparison for lactic acid, as glycolic acid has similar basic spectroscopic features.⁶⁸

Table 2. List of Select SsC Conformer Vibrational Modes Assigned in the IR Spectra with Support from VPT2 Calculated Frequencies, $\tilde{\nu}$, and Intensities, f^a

vibrational mode	$\tilde{\nu}_{\text{calc}}$, cm^{-1} (f)	$\tilde{\nu}_{\text{expt}}$, cm^{-1} (f^b)
ν_1 (OH str. acid)	3559 (49)	3583 (51)
ν_2 (OH str. alcohol)	3543 (15)	3571 (10)
ν_7 (C=O str.)	1763 (100)	1778 (100)
ν_{16} (skeletal mode)	1110 (114)	1124 (197)

^aIntensities are set relative to the intensity of the C=O stretching transition and are included in parentheses. ^bAccuracy of assignments is limited by the 2 cm^{-1} spectral resolution and uncertainty in the peak fitting procedure ranging from 0.03 to 1.2 cm^{-1} among all fitted functions. The overall uncertainty is a combination of both sources; see Table S6 of the SI. Some additional uncertainty is introduced for the ν_7 mode due to the overlap with another signal as well as some interference from gas-phase water.

Because the only difference is the absence of the terminal methyl group, the vibrations from the carboxylic acid and α -hydroxy features are expected to be similar in both molecules; see [Section S3 of the SI](#) for a comparison of structures of these molecules. The two fundamental OH stretching transitions in glycolic acid have been observed with jet expansion Raman spectroscopy and were separated by $8 \pm 4 \text{ cm}^{-1}$, with the transition of the acid OH group being the higher energy transition.⁶⁸ This is again in good agreement with the calculations for lactic acid in this work. In work by Havey et al.,^{69,70} who employed IR spectroscopy at higher temperatures ($\sim 365 \text{ K}$), the two OH stretching (acid and alcohol) fundamental transitions as well as the first overtone transitions could not be distinguished.⁶⁹ However, by using cavity ring-down spectroscopy and investigating the third overtone transition,⁷⁰ they could identify two separate vibrational transitions associated with the two OH oscillators. The 1 cm^{-1} separation between the two OH fundamental transitions was inferred from the harmonic and anharmonic frequencies obtained from Birge–Sponer plots that Havey et al.^{69,70} based on the experimental observation of the fundamental, first overtone, and third overtone features belonging to the OH transitions. Glycolic acid is also suggested as a benchmark molecule for computationally calculating the vibrational anharmonicity by Nejad et al.⁶⁸ Therefore, they used a series of computational methods including CCSD(T)/aug-cc-pVTZ, a very accurate electronic structure method, combined with VPT2 to investigate the vibrational transitions of glycolic acid. They calculated a 3 cm^{-1} splitting between the two OH transitions in glycolic acid, which can be compared to the jet expansion experiment value of $8 \pm 4 \text{ cm}^{-1}$ and the overtone spectroscopy value of 1 cm^{-1} .^{68,70} As part of validating our computational method (B3LYP/may-cc-pVTZ with VPT2), we applied it to glycolic acid and compared it to the high level calculations and experimental results by Nejad et al.⁶⁸ as well as the experimental results by Havey et al.,⁷⁰ see [Section S4 of the SI](#). In general, the VPT2 calculations with the B3LYP/may-cc-pVTZ level of theory employed here predict the transitions to be up to 42 cm^{-1} lower in energy for glycolic acid than the higher level method Nejad et al. used.⁶⁸ This is in agreement with our observations for lactic acid, where the calculations underestimate the transition wavenumbers between 1 and 36 cm^{-1} compared with the experimental results. Our calculations appear to be in good agreement with the work of Nejad et al.⁶⁸ and Havey et al.,⁷⁰ which increases our confidence in the VPT2 results for lactic acid. In summary, the two OH stretch (acid and alcohol) transitions identified theoretically can be assigned to the spectral feature centered around 3580 cm^{-1} , with the acid stretch transition being higher in energy than the alcohol stretch transition, although these are indistinguishable in our experiments without theoretical support.

To fit the C=O signal in our IR spectra, we had to use two Gaussian functions. The main fitted function assigned to the C=O stretch transition is located at 1780 cm^{-1} , whereas a second, very broad signal has a maximum at 1728 cm^{-1} . The origins of this broad band are uncertain, but several factors may contribute. Signals from water ice are routinely observed due to a thin layer of ice that forms on the window to the liquid-nitrogen-cooled MCT detector; see [Section S6 of the SI](#) for details. Furthermore, gas-phase water outside of the cell contributes to a “jagged” look of the broad underlying signal after spectral smoothing. The maximum of this peak varies between experiments, suggesting that it is influenced by

external factors that are difficult to control. Another possible contribution is predicted by the VPT2 calculations, which suggest that two combination bands exist at 1753 and 1747 cm^{-1} with a combined intensity of $\sim 20\%$ of the C=O stretching transition signal; see [Section S7 of the SI](#) for details. Any of these potential sources for the second band might also explain why this band appears broader than other bands observed in the spectrum. A less intense third band is observed near the dominant C=O stretch band at 1837 cm^{-1} . This weaker band is about 1 to 2% of the intensity of the dominant band and is blue-shifted $50\text{--}60 \text{ cm}^{-1}$ from the dominant band. Possible candidates to assign this spectral feature to are the GskC, G'sk'C, and AaT conformers, which have their C=O stretch transition predicted to be 18, 27, and 37 cm^{-1} higher in energy than the SsC C=O stretch transition. The predicted relative populations of the GskC, G'sk'C, and AaT conformers are 2.8, 2.0, and 1.1%, and the calculated intensities are 81, 58, and 19% relative to the calculated intensity of the SsC conformer C=O stretch transition. If we assume the GskC, G'sk'C, and AaT conformers contribute to a combined band, then a signal intensity of about 2 to 5% of the intensity of the SsC C=O stretch transitions is theoretically expected, matching well with the observed signal. Note that both the peak placement and the intensity of this transition are uncertain because this small band was only possible to observe due to the baseline smoothing procedure, which removed systematic noise in our spectra. The C=O stretch transitions in the GskC and AaT conformers were observed by Borba et al.,⁵⁹ who used matrix-isolation spectroscopy with argon as a host gas, to be located at 1781 and 1802 cm^{-1} , respectively, which is higher in energy than the corresponding transition in the SsC conformer transition at $\sim 1766 \text{ cm}^{-1}$. This relative peak placement is in agreement with our VPT2 calculations, which place the C=O stretch transition at 1763 , 1781 , and 1800 cm^{-1} for the SsC, GskC, and AaT conformers, respectively.

With regards to the C–H stretch region ($2800\text{--}3100 \text{ cm}^{-1}$) and fingerprint region ($1000\text{--}1600 \text{ cm}^{-1}$), several different modes are intense enough to be identified, especially the extremely strong 1124 cm^{-1} feature. Assignments in these regions are discussed in [Section S7 of the SI](#), as some of these bands are assigned with less certainty as well as less relevancy to the overall topic of this work.

We also investigated lactic acid deposition onto the CaF₂ windows to rule out any contributions to the gas-phase spectrum by materials deposited on the windows. To do this, we recorded a spectrum of the windows after recording the gas-phase spectrum. A comparison of the IR spectrum of the windows and the gas-phase spectrum can be seen in [Section S8 of the SI](#). We found that there are no observable signals in the gas-phase spectrum belonging to material on the cell windows and could therefore dismiss that as a potential source of spectral interference.

The elevated temperature spectra verified that all of the assigned peaks in [Table 2](#) belong to the same species, the SsC conformer of lactic acid, as the intensities of the assigned signals were observed to increase proportionally with increased temperature. Given how a temperature increase should result in a relative increase in the population of the GskC, G'sk'C, and AaT conformers as well as generally bring more lactic acid into the gas phase, we attempted to find signals in the elevated temperature spectra belonging to these conformers. However, in our experiments, no C=O stretch signals were observed at

higher wavenumbers than the SsC C=O stretch, probably due to the increased noise in the heated spectra. No other features in the elevated temperature spectra could be assigned to higher energy conformers either.

Noncovalent Interactions in Lactic Acid Monomers.

As part of the computational investigation into the different lactic acid conformers, a topological analysis of the calculated electron density was employed in an attempt to quantify the strength of the internal hydrogen bond in the different conformers of lactic acid. AIM analysis did not identify a BCP belonging to the internal hydrogen bond of any lactic acid conformers, except for AaT, which has the carboxylic acid group as the donor and the alcohol group as the acceptor. This is in agreement with work on methyl lactate, which also found no BCPs belonging to internal hydrogen bonding.¹⁷ Because NCI theory provides a regional description of the electron density, it is able to describe the strength of a hydrogen bond in the absence of a BCP using $G(s_{0.5})$. For the isosurface belonging to the alcohol group in lactic acid of the four conformers (SsC, GskC, G'sk'C, SsT), in which the alcohol group is a hydrogen-bond donor, the $G(s_{0.5})$ values were obtained. Figure 3 shows the internal hydrogen $s_{0.5}$ isosurface

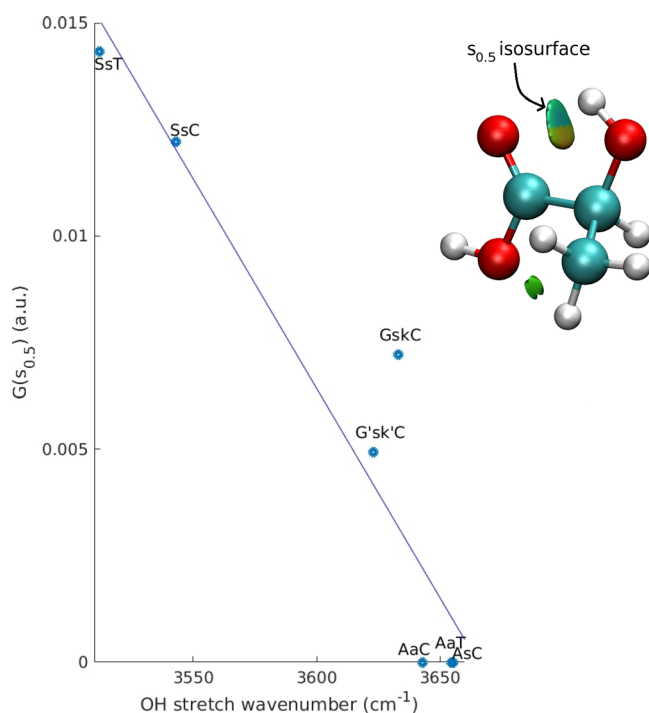


Figure 3. Correlation between the $G(s_{0.5})$ value and the OH stretching frequency for the alcohol OH group in each of the lactic acid conformers. Note that the AaC, AaT, and AsC conformers do not have an internal hydrogen bond with the alcohol group as a donor; therefore, the $G(s_{0.5})$ value is set to zero. The NCI isosurface for the SsC conformer is shown as an example.

of the SsC conformer. The literature indicates that this measure is correlated with the vibrational transition frequency of the hydrogen-bonded mode, both for internal hydrogen bonds^{18,19} and for bimolecular hydrogen-bonded complexes.¹³ In Figure 3, it is seen that a correlation can be made for the lactic acid conformer calculated transition wavenumbers and the $G(s_{0.5})$ value.

The SsT conformer (see the geometry in Figure 1) has the strongest internal hydrogen bond with the alcohol group as the

donor, both in terms of calculated $G(s_{0.5})$ value and in terms of having the lowest vibrational transition frequency, but it is the highest energy lactic acid conformer identified in this work; see Table 1. The SsT and AaT conformers both have a trans orientation of their carboxylic acid group, and this is why they are generally less stable than cis conformers. It is known that for formic acid the two possible conformers (*cis*- and *trans*-formic acid) are separated in energy by ~ 16 kJ/mol.^{71,72} In comparison, the two trans conformers of lactic acid, AaT and SsT, are calculated to be 11 and 19 kJ/mol higher in Gibbs energy, respectively, than the lowest energy cis conformer, SsC. The internal hydrogen bond in the AaT conformer has the carboxylic acid group as the donor, and it is the strongest internal hydrogen bond among all of the lactic acid conformers with a calculated $G(s_{0.5})$ value of 0.017. A correlation can be made between the $G(s_{0.5})$ values of the internal hydrogen bond and the Gibbs energy for the cis conformers of lactic acid, which indicates that the strength of the hydrogen bond influences the conformational stability to a high degree. However, one should be cautionary with such an interpretation because there are many factors playing a role in the Gibbs energy of any particular geometry. For interest, a figure showcasing this correlation has been added to Section S9 of the SI.

The fact that the internal hydrogen bond in SsC is stronger than it is for the GskC and G'sk'C conformers has also been observed in previous studies on methyl lactate.^{17,60} Furthermore, the OH stretching transition was observed for both the SsC and GskC conformers of methyl lactate, which showed the SsC transition to be at a lower wavenumber, indicating a stronger internal hydrogen bond compared with the GskC conformer, which was supported by NCI analysis.¹⁷ Thus it appears that the two most important factors playing a role in the stability of the lactic acid conformer is the orientation of the carboxylic acid group and the strength of the internal hydrogen bond.

Lactic Acid Dimer Structure and Vibrational Spectroscopy Predictions. The lactic acid dimer could potentially be formed in the gas phase like formic acid dimers, which are known to form in the gas phase.^{71,72} The different investigated lactic acid dimer conformers are shown here in Figure 4, and in

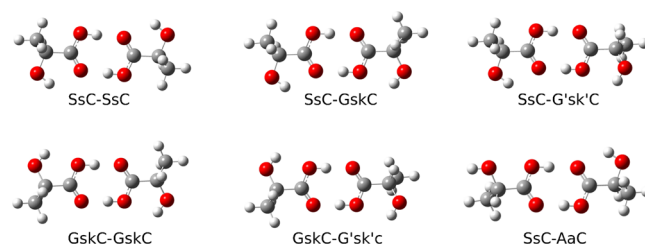


Figure 4. Different lactic acid dimer conformers investigated in this work.

Table 3, the relative Gibbs energies are listed along with the corresponding calculated population. To simplify the discussion, only conformers within 20 kJ/mol Gibbs energy from the lowest energy conformer are included here. The (SsC)₂ dimer was found to be the lowest energy conformer by a significant margin, and thus dimerization is a straightforward process



Table 3. Relative Gibbs Energy (ΔG) between the Lactic Acid Dimers with the Relative Population Calculated at 18 °C along with the Calculated Sum of $G(s_{0.5})$ Values and the Sum of ρ_{HB} Values Belonging to the Intermolecular Hydrogen-Bond Interactions between Lactic Acid and Water

lactic acid dimer	ΔG (kJ/mol)	population (%)	$G(s_{0.5})$ (au) ^a	ρ_{HB} (au) ^b
(SsC) ₂	0.0	86.9	0.023	0.101
SsC–GskC	6.4	6.1	0.025	0.107
SsC–G'sk'C	6.5	6.0	0.025	0.107
(GskC) ₂	12.6	0.5	0.027	0.113
GskC–G'sk'C	12.7	0.5	0.027	0.113
SsC–AaC	16.9	0.1	0.024	0.104

^aSum of the calculated $G(s_{0.5})$ values of the intermolecular hydrogen bonds in the lactic acid/water complexes. See the [Computational Methods](#) section for details. ^bSum of the electron density (ρ) at the bond critical points (BCPs) of the intermolecular hydrogen bonds in the lactic acid/water complexes.

At room temperature, the calculated equilibrium constant for this process is $K = 8 \times 10^3$ at 18 °C, which is comparable to that of formic acid ($K \approx 3 \times 10^2$).^{71–73} However, this calculated equilibrium constant is exponentially affected by the accuracy of the underlying calculated binding energy, which is notoriously difficult to calculate accurately with computational chemistry methods, so the number reported here is best considered as an estimate of the true equilibrium constant.¹⁴ Other examples of strongly bonded carboxylic acid dimers include acetic acid ($K \approx (1 \text{ to } 2) \times 10^3$),^{73–75} propionic acid ($K \approx 1 \times 10^3$),^{73,74} and octo-, nona-, and decanoic acid ($K \approx 5 \times 10^5$).²⁷ The reason that dimer formation is favorable stems from the orientation of the carboxylic acid group being able to form two very strong intermolecular hydrogen bonds in the cyclic dimer.^{27,71–78} This stability of the dimer is, in part, due to the hydrogen bonds being “resonance-assisted hydrogen bonds”.^{79,80} This also explains why double proton transfer, also known as chattering, can occur in the cyclic dimer of carboxylic acids.⁷² In a molecule such as pyruvic acid, in which the lowest energy monomer has an orientation of the carboxylic acid group like that of the SsT and AaT lactic acid conformers, which is due to internal hydrogen bonding, the formation of the hydrogen-bonded dimer is much less favorable.^{81,82} The equilibrium constant for pyruvic acid dimerization is several orders of magnitude lower than it is for formic acid and lactic acid.^{81–83}

For the DFT calculations involving the dimers, care was taken to ensure that only the L-lactic acid (“S” enantiomer) was used, as this is the form used in the theoretical work. It is potentially important to consider, as the existence of two stereocenters in the dimer makes it possible to have chemically distinct diastereoisomers.

The intermolecular hydrogen-bonding strength in the lactic acid dimers depends on which lactic acid conformer is participating in the hydrogen bonding. The GskC and G'sk'C conformers appear to form slightly stronger complexes than the SsC conformer; see [Figure 5](#) and [Table 3](#). Similar results were obtained for complexes of lactic acid with water, as presented in [Table 4](#). This could explain why the energy gap between the SsC conformer and the GskC and G'sk'C conformers slightly narrows when forming the water complex or dimers compared with the monomeric case. The effect behind this is called “hydrogen-bond cooperativity”, where the

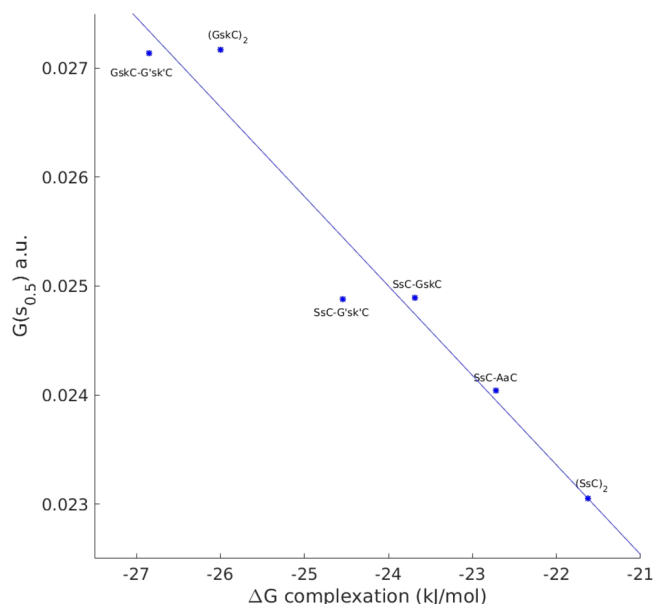


Figure 5. Plot of the summed $G(s_{0.5})$ values of the intermolecular hydrogen bonds in the listed lactic acid dimers versus the Gibbs energy of complexation.

formation of one hydrogen bond affects the formation of a second hydrogen bond.^{84,85} This effect can be positive, by either making the hydrogen-bond donor more electron deficient or making the hydrogen-bond acceptor more electron-rich, or it can be negative, by either making the hydrogen-bond donor more electron-rich or making the hydrogen-bond acceptor less electron-rich. For the GskC and G'sk'C conformers, where the OH group in the carboxylic acid accepts a hydrogen bond from the alcohol, it makes the acid OH group a better hydrogen-bond donor relative to the isolated case. This effect has previously been demonstrated in small water clusters where the hydrogen-bonded mode(s) red shifts with each successive addition of water to the cluster.^{86,87} On the contrary, we observe negative hydrogen-bond cooperativity for the SsC conformer, where the C=O hydrogen-bond acceptor coordinates to two hydrogen-bond donors. In this case, going from a single hydrogen bond to two results in a lower hydrogen-bond strength than would be expected if they were simply additive.⁸⁴ This effect is seen for the lactic acid dimers in the calculated $G(s_{0.5})$ values for the dimers; see [Table 3](#). The SsC dimer displays hydrogen-bond anticooperativity, whereas the GskC and G'sk'C dimers display hydrogen-bond cooperativity. This explains why the $G(s_{0.5})$ value grows when going from the (SsC)₂ conformer with two anticooperative hydrogen-bonding patterns to the SsC–GskC and SsC–G'sk'C conformers, which have one anticooperative and one cooperative hydrogen-bonding pattern, to the (GskC)₂ and GskC–G'sk'C conformers, which have two cooperative hydrogen-bonding patterns. The SsC does, however, remain the dominant conformer among the lactic acid dimers.

On the basis of the large equilibrium constant ($K = 8 \times 10^3$) of lactic acid dimerization, we can estimate the amount of dimer present in the experiments. The vapor pressure of lactic acid is 0.04 Torr at $T = 18$ °C.^{20,21} On the basis of this, the calculated pressure of the dimer is 16 mTorr, or ~40% of the lactic acid monomer concentration in our experiments. Whereas this is a large fraction, we again remind the reader

Table 4. Relative Gibbs Energy (ΔG) between Different Conformers of Lactic Acid/ H_2O (1:1) Complexes with the Relative Population (pop.) Calculated at 18 °C, along with Calculated Sum of $G(s_{0.5})$ Values and Sum of ρ_{HB} Values Belonging to the Intermolecular Hydrogen-Bond Interactions between Lactic Acid and Water

lactic acid/ H_2O complex	ΔG (kJ/mol)	pop. (%)	$G(s_{0.5})$ (au) ^a	ρ_{HB} (au) ^b	$G(s_{0.5})_{\text{complex}} - G(s_{0.5})_{\text{mono}}$ (intramolecular) (au) ^c
SsC- H_2O acid	0.0	93.3	1.9×10^{-2}	6.2×10^{-2}	9.9×10^{-4}
GskC- H_2O acid	8.1	3.4	2.0×10^{-2}	6.5×10^{-2}	2.1×10^{-3}
G'sk'C- H_2O acid	8.2	3.2	2.0×10^{-2}	6.5×10^{-2}	1.9×10^{-3}
AsC- H_2O acid	18.9	0.04	2.0×10^{-2}	7.4×10^{-2}	
SsT- H_2O acid	19.3	0.03	6.3×10^{-3}	3.3×10^{-2}	9.3×10^{-4}
AaC- H_2O acid	19.5	0.03	2.0×10^{-2}	6.3×10^{-2}	
GskC- H_2O alc.	22.6	0.008	8.5×10^{-3}	4.4×10^{-2}	-4.6×10^{-3}
SsC- H_2O alc.	29.7	0.0005	1.1×10^{-2}	5.6×10^{-2}	-9.8×10^{-3}
AaT- H_2O acid	30.4	0.0003	1.5×10^{-2}	6.5×10^{-2}	-1.2×10^{-2}
SsT- H_2O alc.	33.0	0.0001	1.2×10^{-2}	5.7×10^{-2}	-1.2×10^{-2}

^aSum of the calculated $G(s_{0.5})$ values of the intermolecular hydrogen bonds in the lactic acid/water complexes. See the [Computational Methods](#) section for details. ^bSum of the electron density (ρ) in the bond critical points (BCPs) of the intermolecular hydrogen bonds in the lactic acid/water complexes. ^cDifference between the intramolecular $G(s_{0.5})$ value of the monomers of lactic acid and their corresponding 1:1 water complexes.

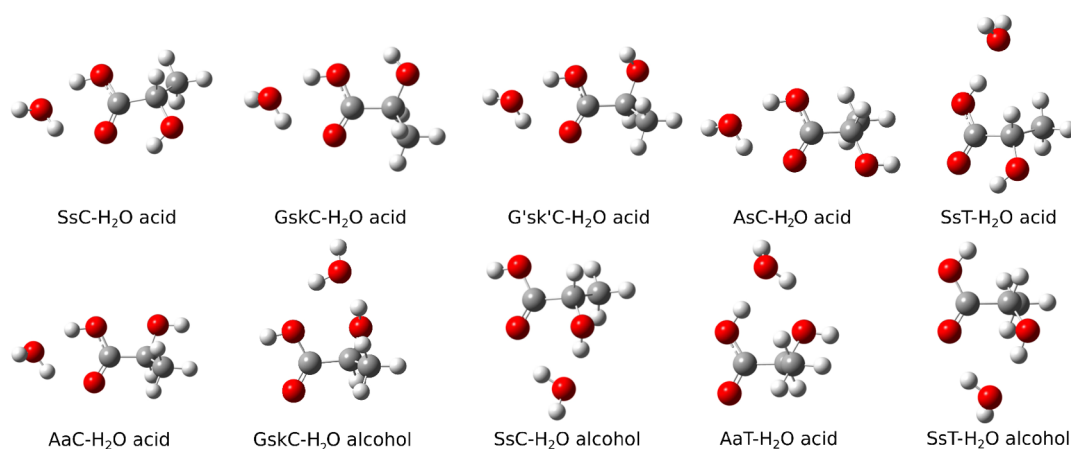


Figure 6. Computed structures of the 1:1 lactic acid/water complex conformers, listed in order of decreasing stability.

that calculations of binding energies and therefore equilibrium constants are notoriously difficult¹⁴ and may be overestimated in this work. The simulated vibrational spectra were used to search for transitions belonging to the dimer, and a list of VPT2 calculated vibrational transitions in the SsC-SsC dimer can be found in [Section S5 of the SI](#). The most intense transition is predicted to be at 2870 cm^{-1} , which is an asymmetric OH stretch of the two acid OH groups. This does coincide with the α -carbon CH stretch transition wavenumber, which we calculate to be centered at 2863 cm^{-1} , which could hinder detection of a dimer signal here. The carboxylic acid dimer OH stretch signals are, in the case of formic acid, known to be smeared out over a broad range, which, if true for lactic acid, could also inhibit detection of the lactic acid dimer because this would spread out the intensity.⁷² This is corroborated by the fact that hydrogen bonding is known to sometimes cause significant broadening of the associated vibrational transition, obscuring spectral detection.^{15,70,88–91} The asymmetric C=O stretch in carboxylic acid is predicted to be red-shifted by 61 cm^{-1} and intensity-enhanced by a factor of 2.2 compared with the monomer, which is consistent with what is seen for other organic acids.^{27,92} It might therefore be possible to observe the dimer C=O stretch signal, perhaps as a shoulder on the monomer C=O signal. The best candidate for this is the very broad spectral feature centered at 1728 cm^{-1} , but we are unable to confidently assign this.

In the elevated temperature spectra, a significant feature was observed at 1245 cm^{-1} . This cannot be assigned to the SsC conformer because it behaves drastically different with temperature compared with all of the other signals assigned to SsC. Two other signals, one at 1378 cm^{-1} and one at 1057 cm^{-1} , show a similar temperature behavior as the 1245 cm^{-1} feature. In a matrix-isolation study, Perera et al.⁶⁴ observed the formation of lactic acid aggregates and rather intense bands from aggregates in the 1200–1300 cm^{-1} region. Thus it is suspected that the 1245 cm^{-1} feature arises from lactic acid condensing onto the windows of the cell in our experiments. The lactic acid film spectrum in [Section S8 of the SI](#) has transitions approximately in those positions, adding credibility to this interpretation. Additionally, Perera et al.⁶⁴ observed that C=O stretch transitions from the aggregates were slightly red-shifted compared with the monomer signal, which also matches the fact that the C=O stretch peak slightly red-shifts in our elevated temperature spectra. As a final point, lactic acid's saturated vapor pressure at 18 °C is ~ 0.04 Torr^{20,21,72} which was enough to form a ~ 0.4 mTorr lactic acid dimer, which could affect the chemistry of lactic acid in the gas phase. However, in an indoor environment like what Pagonis et al.³ studied, the measured lactic acid peak concentration was ~ 7 ppb in the gas phase, which is $\sim 4 \times 10^{-6}$ Torr, calculated at atmospheric pressure in Boulder, Colorado (0.83 bar). Under such circumstances, the dimer pressure would be $\sim 2 \times 10^{-10}$

Torr. In the specific case Pagonis et al.³ studied, there was a significant exchange of indoor air with outdoor air. In small homes or apartments where people spend a lot of time with little air exchange, the lactic acid concentrations can presumably become higher if the exchange of air is low.

Lactic Acid/Water Complexes. The intramolecular hydrogen bonds in lactic acid have been discussed in the section about the lactic acid monomers. Here we examine the possibility of lactic acid forming hydrogen-bonded complexes to water and examine the intermolecular hydrogen bonds using theory. In the experiment presented, the samples used have water as the main impurity, opening up the possibility of forming lactic acid/water complexes. Furthermore, in natural environments, lactic acid is expected to exist alongside water, and as such, gas-phase lactic acid would form complexes with water. Complex formation could influence the chemical processing of lactic acid.^{88,89,93–95} The identified conformers of the lactic acid/water 1:1 complex can be seen in Figure 6, listed in order of relative stability.

Relative calculated Gibbs energies indicate no significant change to the conformer distribution among the carboxylic acid hydrogen-bonded water complexes (see Table 4) as compared with lactic acid monomers. The fact that the acid-bonded water complexes have relative Gibbs energies that resemble those of their corresponding monomers is likely because no significant overall change to their geometry happens upon hydrogen bonding to water. The exception is the AaT conformer, which already had an internal hydrogen bond in the monomeric form, and thus the disruption of this bond can explain the relative increase in energy. The calculated $G(s_{0.5})$ values and ρ_{HB} (HB = hydrogen bond) values in Table 4 are indicators of the overall strength of the hydrogen-bonded complexes. These two values are expected to be somewhat correlated but not proportional, with the $G(s_{0.5})$ values likely being the better measure of hydrogen-bond strength.^{13,19,25} The lactic acid complexes with water coordinated to the carboxylic group of the SsC, GskC, G'sk'C, AaC, and AsC conformers all have nearly identical intermolecular bond strengths based on the $G(s_{0.5})$ values. From the calculations, it is seen that the alcohol-bonded lactic acid/water complexes are all higher in energy relative to their acid-bonded counterparts. This is also reflected in the $G(s_{0.5})$ and ρ_{HB} values listed in Table 4. The change in $G(s_{0.5})$ value for the internal hydrogen bond, $G(s_{0.5})_{\text{complex}} - G(s_{0.5})_{\text{monomer}}$ (see Table 4), reflects whether the internal hydrogen bond in the monomers is weaker or stronger than the corresponding 1:1 lactic acid/water complex. This shows that water coordinating to the free OH group in each lactic acid conformer is associated with an enhanced strength of the internal hydrogen bond as a derived effect from complexation. On the contrary, coordinating to the OH group participating in internal hydrogen bonding will weaken the internal hydrogen bond. It can also be deduced that in all cases, coordination to the free OH group is energetically more favorable than coordinating to the internally hydrogen-bonded OH group.

In comparison, for pyruvic acid, the conformer distribution notably changes upon forming the water complex or when making the pyruvic acid dimer.⁸² This is due to the fact that the lowest energy conformer of pyruvic acid has the carboxylic acid group forming an internal hydrogen bond with the ketonic C=O group as the acceptor, which is disrupted when forming water complexes. On the contrary, the second lowest energy conformer of pyruvic acid has the same orientation of the

carboxylic acid group that the lowest energy conformer of formic acid and lactic acid has, which causes it to readily form intermolecular hydrogen bonds with little change to its geometry.^{82,83} The equilibrium constant for forming the 1:1 lactic acid/water complex is calculated to be $K = 8$ ($T = 18$ °C). The vapor pressure of lactic acid at laboratory temperature, $T = 18$ °C, is 0.04 Torr.^{20,21} Given that the gas-phase water features were barely observable in the spectrum, we did not expect any formation of lactic acid/water complexes in our experiments. In the context of indoor environments where the relative humidity likely falls within the 10–90% range, the relevance of the lactic acid complexes with water is potentially much higher. Water, as the third most abundant molecule in the atmosphere, is known to significantly impact gas-phase chemistry,⁹⁶ including its interaction with organic acids,¹⁰ making it worthwhile to consider in future studies.

CONCLUSIONS

We investigated lactic acid in the gas phase and examined its ability to form intra- and intermolecular hydrogen bonds. IR spectra of gas-phase lactic acid were successfully recorded with FTIR, despite experimental challenges due to low vapor pressure, and multiple IR features were assigned to the dominant conformer (SsC) with help from computational chemistry. The employed DFT method and basis set combination, B3LYP/may-cc-pVTZ, along with VPT2 analysis was able to reasonably predict the vibrational transition wavenumbers and relative intensities. Additional coupled-cluster single-point calculations were used to obtain more accurate electronic energies, which improved our confidence in the calculated relative energies of conformers and complexation energies. Subsequent electron density topology analysis of the internal hydrogen bonds in lactic acid unveiled that the strength of the internal hydrogen bond was one of the two most important predictors for the relative energy between conformers. The other main predictor for relative energy is the orientation of the carboxylic acid group, with a cis conformation being preferred over the trans conformation by 11 to 19 kJ/mol in Gibbs energy.

Lactic acid complexation with water and lactic acid dimerization were also investigated in this work. Spectral features belonging to the lactic acid/water 1:1 complex as well as the lactic acid hydrogen-bonded dimer were sought after in the spectra, but no signals could be reasonably assigned to these complexes. The overall distribution of conformers in the aforementioned complexes was theoretically found to resemble that of the lactic acid monomer. This was explained using electron density topology analysis, which found that the different conformers generally formed intermolecular hydrogen-bonded complexes of similar strengths, and therefore the overall conformer distribution was mostly unchanged.

Lactic acid in indoor environments exists with different molecules, including other carboxylic acids, emitted from humans or human activity.³ This means the extent of hydrogen-bonded complex formation involving lactic acid exceeds the two cases studied here, dimerization and complexation to water, and could contribute to chemistry indoors.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpca.0c09341>.

¹H NMR analysis of the lactic acid sample. Details of the sample cell used for gas-phase spectra collection. Overview of molecular structures with the conformer naming convention. Additional validation of the employed computational methods. List of calculated anharmonically corrected frequencies and intensities for the lowest energy conformer of lactic acid, the lactic acid/water 1:1 complex, and the lactic acid dimer. Details of spectral interference from water ice. Overview of Gaussian functions fitted to the experimental spectrum and compared with the calculated spectrum. Comparison between the gas-phase lactic acid spectrum and a spectrum of the CaF₂ windows after running the gas-phase measurements. Supplemental discussion about NCI results. Additional details of the computational methods employed including the comparison of results obtained by different methods (PDF)

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Notes

The authors declare no competing financial interest.

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